

Impact of Statistical LCoS-induced PDL in WDM Networks Equipped with Multirail Amplifiers and 800ZR+ Interfaces

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Abstract *This study quantifies the benefit on both the OSNR penalty and the network capacity of accounting realistically statistical PDL insertions vs. fixed reasonable worst-case values in the context of modern optical networks equipped with multirail amplifiers featuring integrated spectral equalizers.*
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Introduction

After the fruitful introduction of the first power-saving Coherent Pluggable Interfaces (CPI) [1-4] six years ago, our industry is now revamping the WDM amplifiers to also make them much more power efficient and compact. This new "multirail" version better integrates several parallel amplifiers sharing more functionalities [5]. One of them is the internal Digital Gain Equalizer (DGE) to keep very flat the spectral profile of the WDM comb outputting from the amplifier under any operating condition. Liquid Crystal on Silicon (LCoS) is one of the most likely technologies for these shared DGEs [6]. It also induces physical degradations associated to LCoS, like Polarization Dependent Loss (PDL) similar to the one of the LCoS-based Wavelength Selective Switch (WSS) [7,8]. Hence, deploying these MultiRail Amplifiers (MRA) at scale in large transparent WDM networks will imply more frequent PDL contributions to the transmitted signal than WSS, as WDM channels usually traverse more in-line optical amplifiers than routing WSS's. On this basis, this study examines the impact of MRA on the global performance of two core WDM networks equipped with 800ZR+ CPIs [9], depending on whether its PDL-associated distortion on transmitted channels is modelled from statistical PDL values or from a reasonable worst-case value.

Penalty on 800ZR+ channels due to PDL

To derive the PDL-induced penalty on the Signal to Noise Ratio (SNR) we resorted to enhanced Gaussian Noise (GN) model extended to PDL [10] in order to compute the SNR distribution due to PDL. Along the same lines of [11], the SNR performance of the two polarizations is converted into Bit Error Rate (BER) and the overall SNR performance is inferred by the polarization-averaged BER [11, Eq. 1]. The availability of the SNR distribution allows to assess a level of risk usually called out of service probability [12, Fig. 5] or Outage Probability (OP) [11]. We hence derived the PDL-induced penalty on the SNR in presence of distributed Amplified Spontaneous Emission (ASE) noise considering a target OP of 10^{-5} and neglecting the Kerr Effect. The penalty expresses

the increase needed by the reference Optical SNR (OSNR) as compared to the one in absence of PDL, to ensure a transmission characterized by an OP below the desired target. This result is achieved if the area of the tail of the SNR distribution falling below the SNR value at the limit of the Forward Error Correction (FEC), i.e. the cumulative distribution function at the SNR FEC value, is lower than the targeted OP. In our study, we focused on a target OP of 10^{-5} and the OSNR reference in absence of PDL was relative to the FEC value plus an extra OSNR margin of either 0.5 dB or 1 dB. The PDL contributions experienced by a signal traversing the network add up in a stochastic way given the random alignment of axes of crossed elements introducing PDL, hence the cumulative PDL is itself a random variable that is usually characterized by its Root Mean Square (RMS) value [13].

Fig. 1 shows the derived PDL-induced penalties vs the PDL RMS value, for four of the channel modulations of the 800ZR+ CPIs [1]. Solid (resp. dashed) lines refer to the penalty considering an extra margin of 1 (resp. 0.5) dB on the reference OSNR. The penalties shown on Fig. 1 are computed with the simulation of a reference cascade of PDL elements representing the network elements introducing significant PDL, as regular Erbium Doped Fiber Amplifiers (EDFAs) and WSSs. Different PDL RMS values are obtained considering different numbers of repetitions of

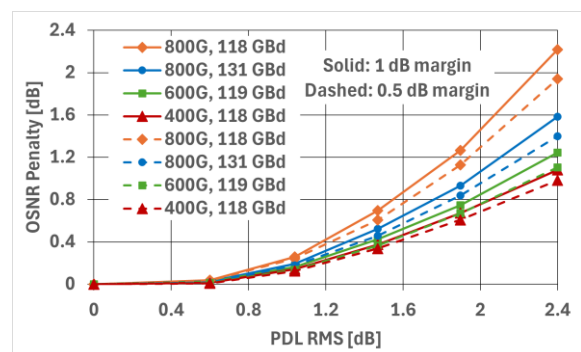


Fig. 1: Penalty vs PDL RMS value for four channel modulations of the 800ZR+ OpenROADM standard considering uniformly distributed ASE noise, with 0.5 dB (dashed lines) and 1 dB (solid lines) extra margin on the reference OSNR.

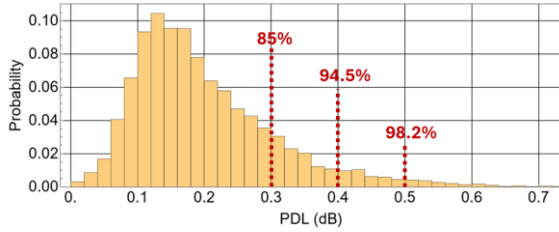


Fig. 2: Distribution & percentiles of 20 million random draws of LCoS-based WSS PDL as measured in [7].

the reference cascade. By this way a larger number of repetitions represents a larger transmission distance. A common approach when dealing with the PDL brought by WSSs is to assume a reasonable and fixed amount of PDL introduced by each WSS. However, it has been shown [8] that a more realistic way to account for the introduced PDL should use a statistical approach. In our study we will compare both approaches, by using the distribution depicted in Fig. 2 for a more realistic accounting for the PDL introduced by any LCoS-based network element, i.e., either WSS, DGE, and MRA.

It is worth noting that a stochastic PDL of LCoS-based elements translates into a variable PDL RMS for a given number of cascaded elements. Hence, unlike the common approach leading to penalties depicted in Fig. 1, now a given PDL RMS value could be obtained by very different numbers of cascaded PDL elements. From a network simulation standpoint, it questions also the way of accounting for the actual PDL-induced penalties since results like those of Fig. 1 should be ideally needed for each PDL cascade including random draws from Fig. 2 distribution. Since such a comprehensive analysis is clearly too cumbersome to be carried out, we investigated if penalties obtained with a fixed PDL for each LCoS-based element are pertinent also for the statistical approach. Hence, we compared the penalty obtained with fixed PDL for each WSS for the 118 GBd 800G format, displayed in Fig. 1, with the penalty obtained by replacing each WSS related PDL with a stochastic value drawn

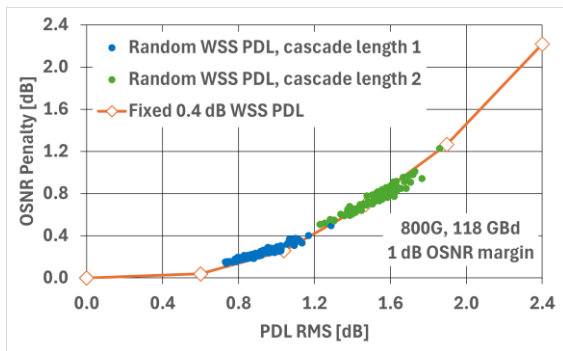


Fig. 3: Penalty vs PDL RMS value for the 118 GBd 800G profile of 800ZR+ CPIs considering distributed ASE noise. Line: fixed 0.4 dB PDL for each WSS. Isolated markers: PDL of each WSS randomly drawn from Fig. 2 distribution.

according to the distribution of Fig. 2. We focused on two different repetitions of the reference cascade, leading to PDL RMS values of 1.5 dB and 2.4 dB in Fig. 1, and we performed 100 random draws for each of these cascades.

Fig. 3 reports with isolated markers the penalty obtained for each random draw, encompassing two different cascade lengths. We observe that the primary effect of the statistical approach is to significantly reduce the PDL RMS value for a given cascade length. This was expected since the median of the distribution of Fig. 2 falls below the 0.4 dB PDL value, that was used in the common approach as reasonable worst-case since it represents the 94.5% percentile of Fig. 2 distribution. We observe that for the longer cascade, the PDL RMS among 100 draws reaches at most 1.85 dB, relative to a penalty of 1.2 dB, 1 dB less than the 2.2 dB penalty at PDL RMS of 2.4 dB from the corresponding common approach. On average, we observed 1.45 (resp. 0.45) dB less penalty for the longer (resp. shorter) cascade in Fig. 3 thanks to the PDL RMS average reduction from 2.4 (resp. 1.5) dB to 1.5 (resp. 0.9) dB. We also observe from Fig. 3 that the penalties obtained with the statistical approach slightly deviate from those expected at the same PDL RMS with a fixed PDL per element (orange line). However, we quantified the maximum misalignment between isolated markers and the reference line, finding 0.1 dB as largest penalty underestimation and 0.14 dB as largest penalty overestimation. Penalty underestimation is the most critical since it leads to consider lower penalties than real ones, hence, to overestimate the actual quality of transmission. Given the very limited amount of penalty underestimation, we conclude that we can resort to penalties highlighted in Fig. 1 also when adopting the statistical approach, by simply considering the PDL RMS value coming from the random draw.

Model of network simulation

We emulate the two network topologies sketched in Fig. 4 to assess the relevance of a statistical model of PDL for network dimensioning. TID30N is a national Spanish backbone network having 30 nodes and 56 links. EBN39 is a core Pan-European network with 39 nodes and 59 links.

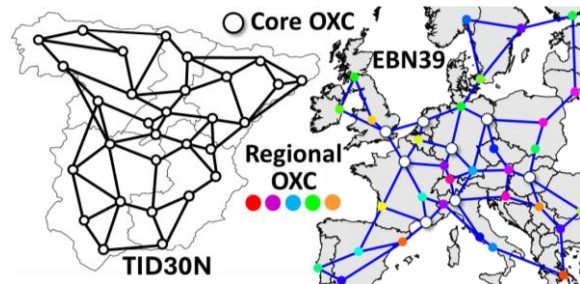


Fig. 4: TID30N [14] and EBN39 WDM networks.

These two topologies are modelled with 16 pairs of bidirectional fibers per link and with 800ZR+ CPI aligned on the four standardized channel modulations presented in previous section. The fiber infrastructure relies on standard single mode fiber with 0.2 dB/km optical loss and 75 km maximal length of fiber span. An in-line amplifier follows each fiber span to offset its loss with 5.5 dB noise figure over the 4800 GHz-wide C-band. It could be a discrete EDFA or part of a MRA with integrated DGE. The routing optical nodes are as per the conventional WSS-based "Route&Select" layout [15]. The C-band transmission bandwidth transports thirty-two 800ZR+ channels 150 GHz apart. Each channel carries its maximum data rate resulting from its SNR along its Light Path (LP). This SNR is established in accordance with the GN model as the inverse of the sum [11] of inversed optical linear and non-linear noises cumulated along the LP plus the inverse of the ceiling of CPI electrical SNR due to CPI imperfections and to the power of the received signal. The PDL impact is approximated by multiplying the final linear noise with the linear penalty induced by PDL. Due to their large size difference, EBN39 and TID30N are simulated differently. EBN39 has optoelectronic regeneration to transmit along LPs even too long for the Quadrature Phase Shift Keying 400 Gb/s CPI format. Moreover, its connections carry from 800 to 3200 Gb/s with 800 Gb/s steps, and their geographical distribution is hierarchical, i.e. 33.3% of them interconnect random pairs of core nodes (see Fig. 4) and 66.6% are between any regional node and its 3 closest core nodes. The smaller TID30N network does not need regeneration. Its traffic has uniform geographical distribution between any pair of nodes. The random capacity of its connections is evenly distributed over the following set {800 Gb/s, 1600 Gb/s, 2400 Gb/s, 3200 Gb/s}. For each individual network simulation, the traffic is randomly drawn with a specific random seed. The PDL of the LCoS-based network elements such as WSS, DGE, MRA is modelled in two distinct ways: either this is a PDL value uniform over the whole network or a randomly selected value per element as per the distribution reported in Fig. 2. For a given network simulation, each connection is allocated one after another if a suitable LP exists with enough free spectrum from its source node to its destination one. When such a LP cannot be found, the related demand is dismissed. Inverse multiplexing and end-to-end traffic grooming are enabled. The simulation stops once more than 1% of the total demanded throughput has been rejected. The total capacity accommodated so far is then called the Max Network Capacity (MNC). At this point, we also defined N_{CPI}

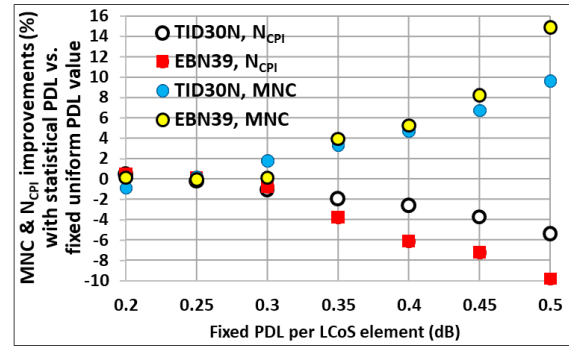


Fig. 5: MNC & N_{CPI} improvements if account for actual PDL values vs. for fixed uniform worst case values.

as the mean number of CPIs allocated per Gb/s transmitted. These MNC and N_{CPI} metrics quantify the network performance and their values reported in the remainder are averaged over 100 random traffic and PDL draws for each simulated network configuration.

Results of network simulations

Our simulation outcomes appear in Fig. 5 comparing the percentages of MNC and N_{CPI} improvements if assuming realistic statistic PDL of DGE and WSS versus a fixed common value representing a reasonable worst-case PDL. The highest gains occur for the largest EBN39 network, as its optical connections go on average through more MRAs than in TID30N. Hence, for EBN39, MNC can enhance up to 15% (and N_{CPI} by nearly -10%) as compared to a very conservative 0.5 dB PDL assumption, while less conservative 0.4 dB PDL assumption brings down the MNC enhancement to around 5% for both EBN39 and TID30N.

Conclusions

This study reports original numerically simulated OSNR penalty induced by PDL on 118 and 131 GBaud standardized OpenROADM channels. We also showed that penalties obtained by assuming conservative fixed PDL contribution by any LCoS-based network element are useful also in the more realistic case of statistical PDL contribution, provided that the correct PDL RMS value is considered for the actual cascade. By focusing on future scenario of transparent WDM networks featuring large bundle of parallel fibers associated with MRA at each fiber span, our simulations show that realistic statistical PDL contribution could bring up to 15% extra network capacity and nearly -10% fewer CPIs per Gb/s than a very conservative approach for the EBN39 topology. Such a performance difference underlines the relevance for next-gen networks of more sophisticated PDL model than simple worst-case values. This model should be able to better reflect the statistical nature of the PDL value in dB of MRAs, as it will likely depend on both DGE technology and setup specific for each DGE vendor.

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